

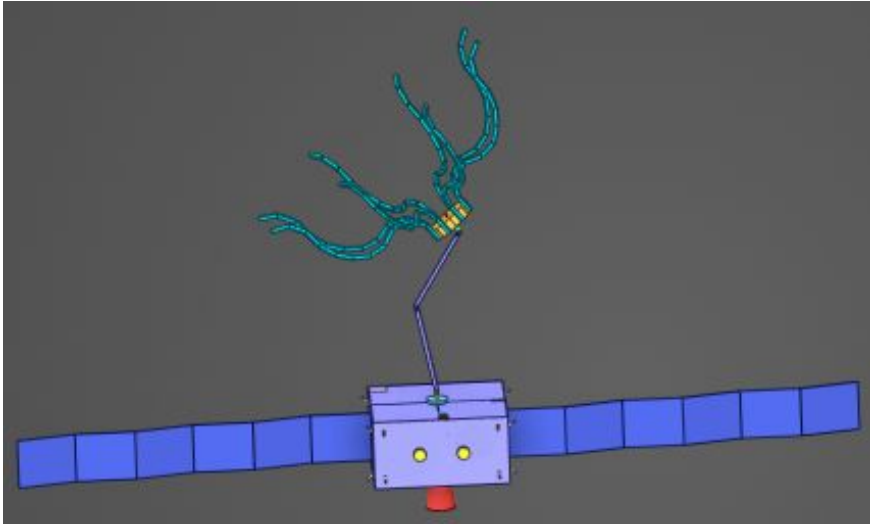
PERSEUS

Protective Environmental Recovery Spacecraft for Expended Upper-Stages



Team Member	Responsibility
Jonathan Acuna	Power Communications & Data Thermal Control (Primary)
Kadin Caldwell	Configuration & Structure (Secondary) Thermal Control (Secondary)
Shea Davis	Payload Space Environment Mechanical
Andrew Mills (Team Lead)	GN&C ADCS Configuration & Structure (Primary) Reliability & Risk Cost
Savannah Romaine	Summary/Overview Launch Vehicle Selection Propulsion Summary & Conclusion
Andrew Scott	Astrodynamics Launch & Mission Operations

Picture of PERSEUS and Target (S/L-16 Zenit-2 upper stage, (SATNO 28353))



Mission Overview/RFP

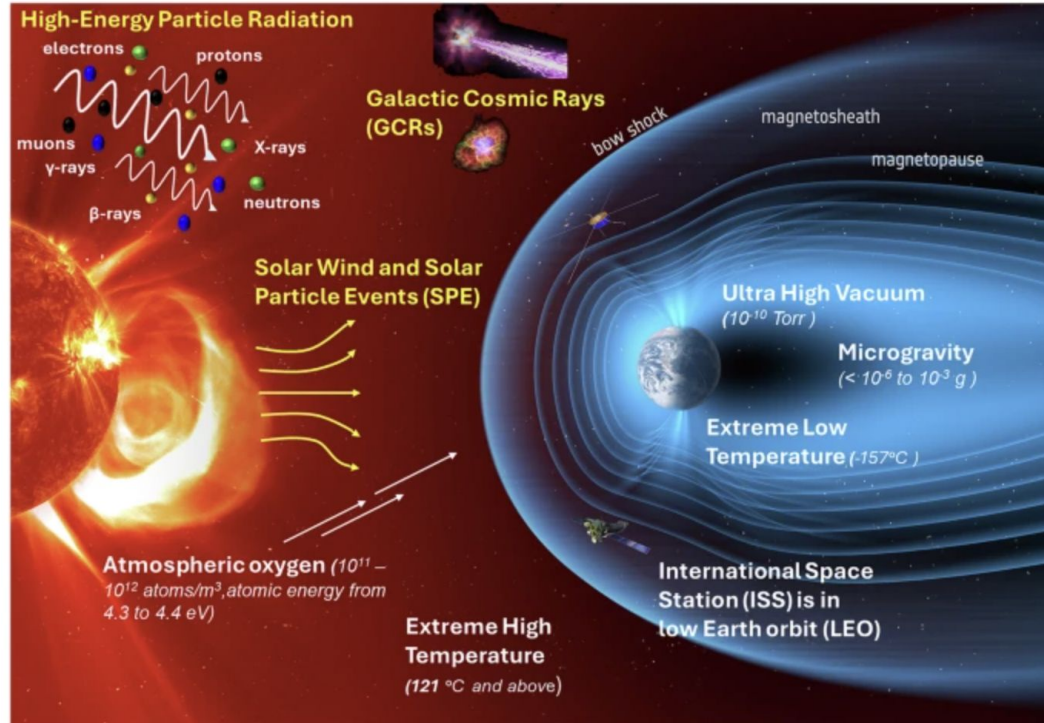
- Goal: Demonstrate the ability to rendezvous, capture, and dispose of a large piece of space debris by means of controlled atmospheric re-entry
 - Close Proximity Operations
 - Debris > 5000 kg
- Mission: Go to LEO to capture a Zenit-2 upper stage using a robotic arm capture mechanism and dispose of the debris by burning up on controlled re-entry

Target Object Payload

- S/L-16 Zenit-2 Upper Stage (SATNO 28353)
- Most concerning object currently in LEO
- Size/Weight: 11.047 m x 3.9 m / 9000 kg
- Spinning Motion: caused by outgassing and residual forces
 - gradually slows due to internal damping & Earth gradient stabilization
- Engine-to-Zenith Orientation



Space Environment Overview: Why It Matters for a 20-Year-Old Booster



Space Environment - The Vacuum Environment

Absence of Atmospheric Pressure Affects:

- Material Outgassing
 - Outgassing = release of volatile compounds from materials
 - Causes mass loss and surface residues
- Thermal Control
 - Cooling relies solely on radiation and conduction
- Structural Integrity

Space Environment - The Neutral Environment

AO Attack

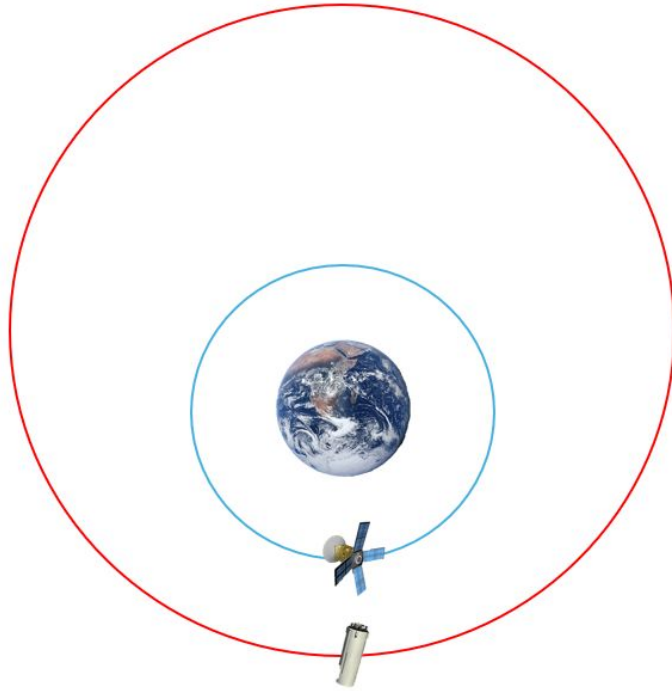
- AO is created from photodissociation of O_2 by solar UV radiation
- AO fluence peaks in the ionosphere
- Energy: ~ 5.2 eV at 7.8 km/s orbital velocity
- Effects:
 - Erosion and oxidation of materials
 - Changes in thermal properties (weakens structural materials)
 - Surface roughening

Space Environment - The Radiation Environment

- Solar Electromagnetic Radiation
- Galactic Cosmic Rays (GCR)
- Solar Particle Events (SPEs)
- Effects:
 - Non-ionizing (UV) radiation: surface degradation
 - Ionizing radiation: semiconductor and material damage
 - Leads to damaged coatings, embrittlement of outer layers

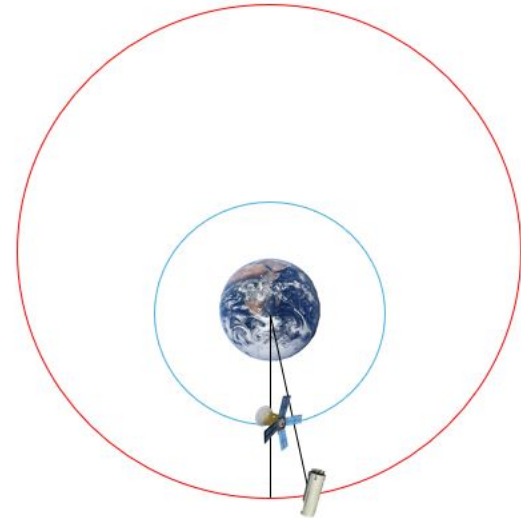
Astrodynamics - Phases

- Ascent From Earth
- Parking Orbit
- Hohmann Transfer
- Characterization
- Rendezvous
- Capture
- Deorbit



Astrodynamics - Ascent from Earth/Parking Orbit

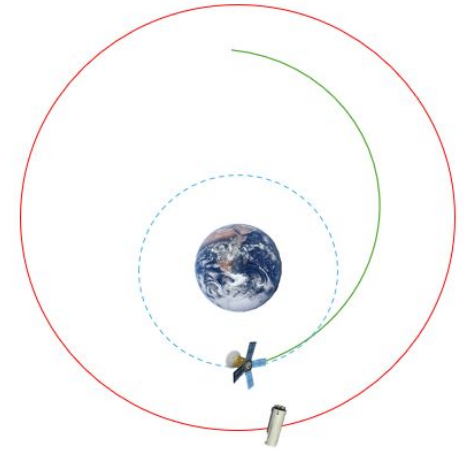
- Launch on Falcon 9 rocket into 743.81 km parking orbit
- Match inclination and RAAN of target
- Test and checkout
- Wait for 1.866 degree lead angle before transfer



Astrodynamics - Hohmann Transfer

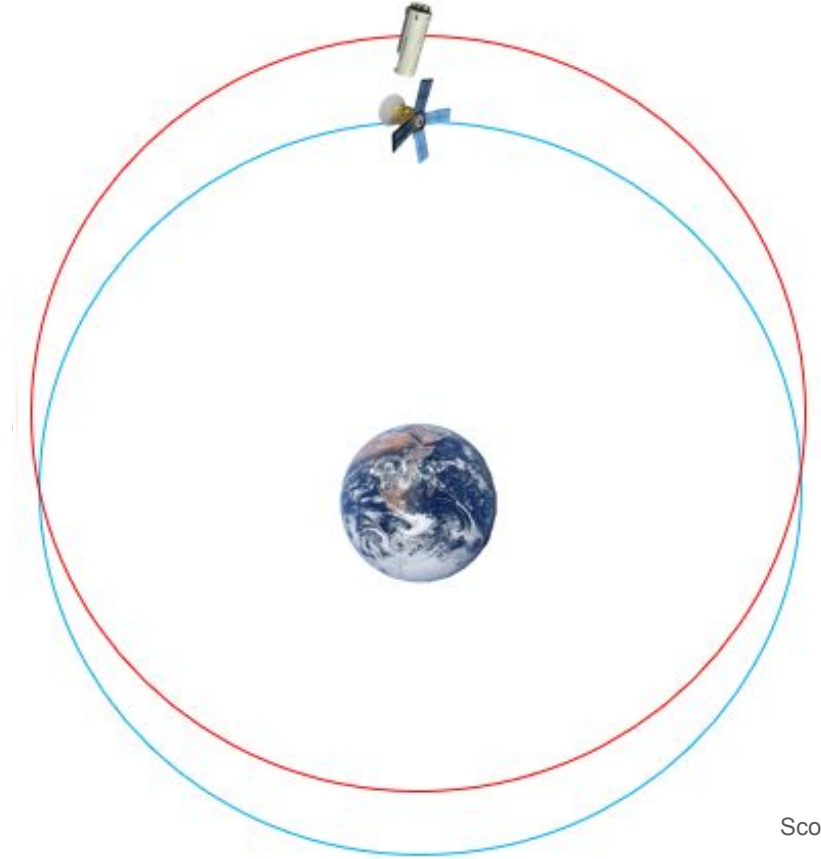
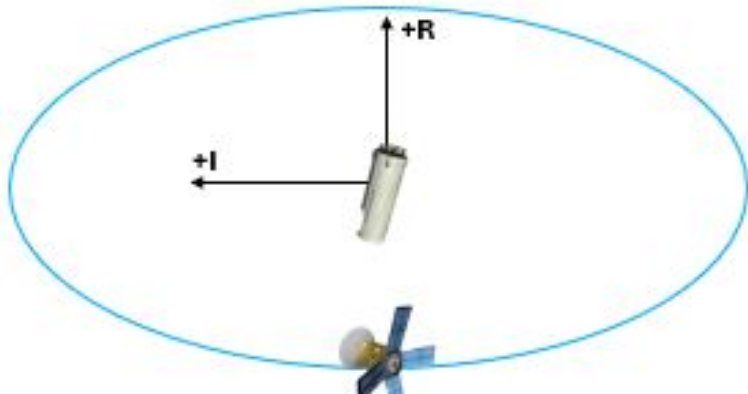
- Hohmann and Edelbaum analyzed - Hohmann selected
- Targeting circular orbit and matching semi major axis
- Arriving at Target's apogee to establish Natural Motion Circumnavigation
- Delta V table below

Event	Velocity Value (km/s)
Initial orbit velocity in parking orbit	7.481171
Velocity of circular target orbit	7.429196
Delta-V to enter Hohmann transfer	0.026032
Delta-V to establish circular target orbit	0.025942
Total Hohmann transfer Delta-V	0.051974



Astrodynamics - Characterization/Rendezvous/Capture

- Natural Motion Circumnavigation
- Utilizes zero fuel
- Characterize all 6 sides of target
- Determine spin rate
- Close with target
- Grab target



Astrodynamics - Deorbit/Reentry

- Hohmann transfer targeting 75 km altitude
- Reentry at 120 km altitude
- Flight path angle of -1.509 degrees
- Touchdown in Pacific Ocean

Key Event	Velocity Value (km/s)
Delta-V to close with target from NMC orbit	0.010726
Delta-V to deorbit	0.213236
Total Delta-V (closing and deorbit)	0.223962
Total Mission Delta-V (Hohmann transfer, closing, and deorbit)	0.275937



Launch Vehicle Selection

- Compared ULA (United Launch Alliance) Vulcan and SpaceX Falcon 9
 - Western Range due to inclination requirements
- Selected Falcon 9
 - Available data and files
 - Cost
 - Reliability

	Falcon 9	Vulcan
Cost	~ \$70 Million	~ \$110 Million
Number of Launches	~ 538	~ 2

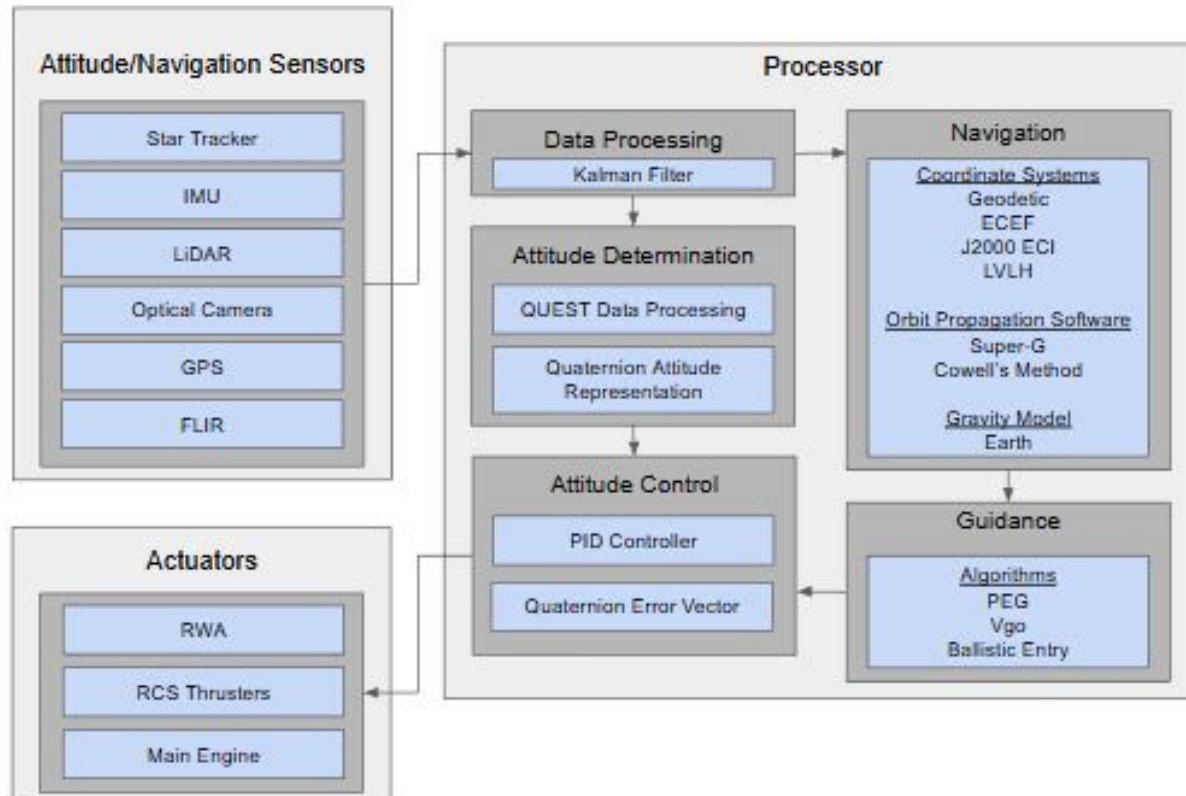
Launch Vehicle Selection

- Compared Circular Payload Attachment Fittings (CPAF) for Falcon 9
 - All fairing envelopes are the same size
- Selected the 2624 mm CPAF
 - Middle size option due to early decision in design process
 - Allowable Mass of 19,050 kg
 - Smallest CPAF allows 10,885 kg

GNC - Navigation Hardware

Instrument	Function
Star tracker (Terma T3 Star Tracker)	Provide accurate pointing and navigation information to correct IMU drift and add redundancy to the GPS system
Inertial Measurement Unit (HBK 3DM-CX5-IMU)	Provide data on angular velocity changes and linear accelerations on all axes
Flash LiDAR (ASC GSFL-16KS)	Provide target-relative navigation during the characterization and docking phases
Optical camera (Infinity Avionics ORION12MP-550)	Provide target-relative navigation and visualization during the characterization and docking phases
Forward Looking Infrared (FLIR) (Teledyne Blackfly S BFS-U3-13Y3)	Provide target-relative navigation from the center of the REACCH end-effector
GPS Receiver (L3Harris AS-48917 GPS Band Omnidirectional Antenna)	Provide navigation data throughout the mission

GNC - System Block Diagram



GNC - Phases

	Ascent	Parking Orbit	Orbit Adjustment	Target Rendezvous	Deorbit
Guidance algorithms	N/A	Vgo, PEG	Vgo, PEG	Vgo, PEG	Vgo, PEG, Ballistic entry
Maneuvers supported	N/A	Momentum Dumping	Hohmann Transfer	Lambert targeting CW relative motion	Hohmann Transfer
Navigation algorithms	N/A	Orbit Determination: State vector and COE determination Coasting: Cowell's method Powered: Super G method	Orbit Determination: State vector and COE determination Coasting: Cowell's method Powered: Super G method	Orbit Determination: State vector and COE determination Coasting: Cowell's method Powered: Super G method	Orbit Determination: State vector and COE determination Coasting: Cowell's method Powered: Super G method

ADCS - Overview

- Various subsystems onboard PERSEUS require pointing, including orienting antennas towards ground stations, pointing solar panels towards the sun, and aligning the grabbing mechanism for capture
- Attitude determination sensors collect pointing data in relation to the J2000 ECI reference frame, and use a Kalman filter to filter through the data.
- This data feeds into the control algorithms and actuators to correct errors between desired and actual attitude

ADCS - Pointing Capabilities

Spacecraft Requirement Item	Requirement Data	Units	Contributing Component
GPS Receiver Targeting Direction	Zenith		GPS Receiver
Spiral Antenna Targeting Direction	Nadir		Spiral Antenna
Star Tracker Targeting Direction	Space, 37.5 deg from sun (full moon accepted)		Star Tracker
Solar Array Targeting Direction	Sun		Solar Array
Capture Mechanism Targeting Direction	Target		Capture Mechanism
Optical Camera Targeting Direction	Target		Optical Camera
Flash LiDAR Targeting Direction	Target		Flash LiDAR
Pointing Control Error	10	Arcsec	RWUs, RCS
Pointing Knowledge Error	10	Arcsec	Star Tracker, IMU
Pointing Jitter	0.1/60	Degrees/sec	Flash LiDAR, Optical Camera
Pointing Knowledge Error Temperature Stability	0.04% for IMU 0.1 for Star Tracker	%/Kelvin, Arcsec/Kelvin	Star Tracker, IMU
Slew Rate	<0.3 full performance <1.5 reduced performance	Degrees/sec	Star Tracker
Slew Rate Resolution	<0.003	Degrees/sec	IMU
Independent Antenna Pointing DOF	0	Degrees	Spiral Antenna
Independent Solar Array Pointing DOF	360	Degrees	Solar Array

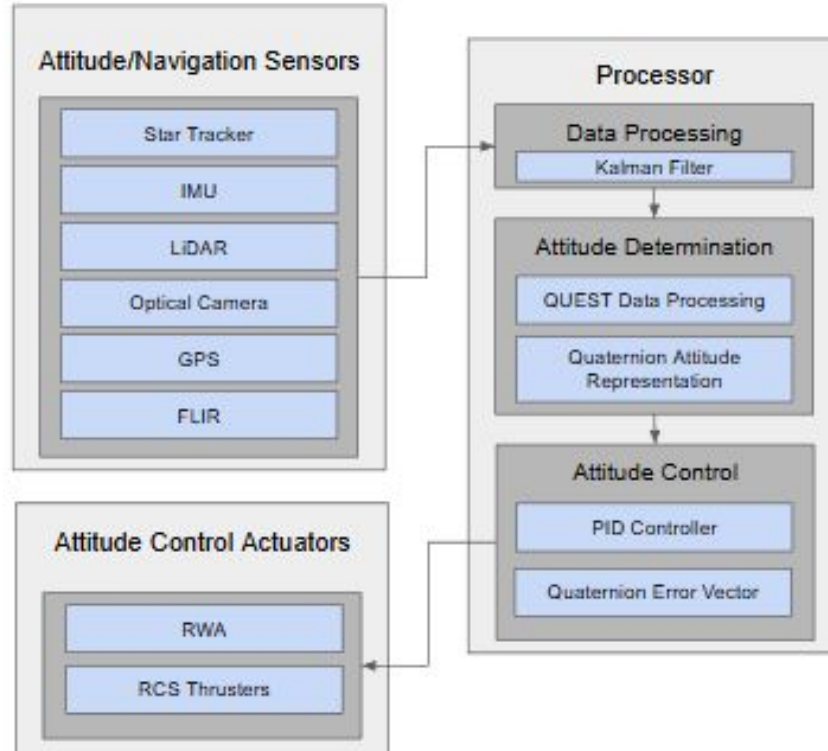
ADCS - Attitude Determination Sensors

Instrument	Function
Star tracker (Terma T3 Star Tracker)	Provide accurate pointing and navigation information and correct IMU drift
Inertial Measurement Unit (HBK 3DM-CX5-IMU)	Provide data on angular velocity changes and linear accelerations on all axes

ADCS - Attitude Control Actuators

Instrument	Function
Reaction Wheel Assembly (Space Inventor WHL-550 mNms)	Use motors and wheels to harness angular momentum in order to change the orientation of a spacecraft along any axis. Periodically requires a momentum dumping maneuver from the RCS thrusters to avoid exceeding momentum design limits.
Reaction Control System Thrusters (Aerojet Rocketdyne MR-106L)	16 thrusters provide quick control to spacecraft roll, pitch, and yaw. Assists with pointing, orbit maintenance, small maneuvers, and momentum dumping.

ADCS - Block Diagram



ADCS - Moment of Inertia

Variable	PERSEUS (Solid Rectangular Prism)	Arm (Solid Cylinder)	Target (Hollow Cylinder)	Combined
Length (m)	1.92			
Width (m)	1.92			
Height (m)	0.98	0.65	11.05	
Diameter (m)		0.61	3.90	
Thickness (assumed) (m)			0.05	
Mass (kg)	1,723.26	90.00	9,000.00	10,813.26
I _x (kg*m ²)	667.30	14.86	108,420.44	175,350.51
I _y (kg*m ²)	667.30	14.86	108,420.44	175,350.51
I _z (kg*m ²)	1,058.77	4.18	33,786.56	34,849.56

ADCS - Environmental torques

- Solar Radiation Pressure Torque: Solar pressure pushes against the sun-facing area of a spacecraft, inducing torque on non-symmetrical areas
- Aerodynamic Torque: Atmospheric constituents push against the ram-facing area of a spacecraft, inducing torque on non-symmetrical areas
- Magnetic Field Torque: Torque induced by Earth's magnetic field
- Gravity Gradient Torque: Torque induced by separated areas of different mass on the spacecraft. This is the most significant torque experienced by

PERSEUS

Disturbance Torque	PERSEUS	Combined Configuration
Solar Radiation Pressure (Nm)	0	1.02E-03
Aerodynamic (Nm)	0	8.70E-04
Magnetic (Nm)	3.97E-05	7.93E-05
Gravity Gradient (Nm)	1.08E-04	3.89E-02

Propulsion - Introduction

Topics Covered:

- Main Spacecraft Engine Selected

- Main Spacecraft Engine Schematic

- Main Spacecraft Engine Tank Sizing Results

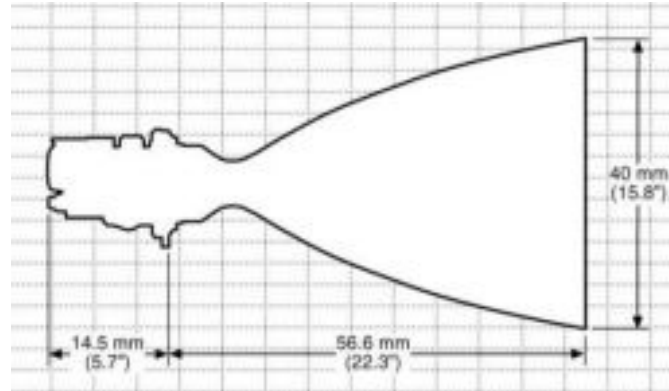
- RCS Thrusters Selected

- RCS Thrusters Schematic

- RCS Thrusters Tank Sizing Results

Propulsion - Main Spacecraft Engine: R-40B Engine

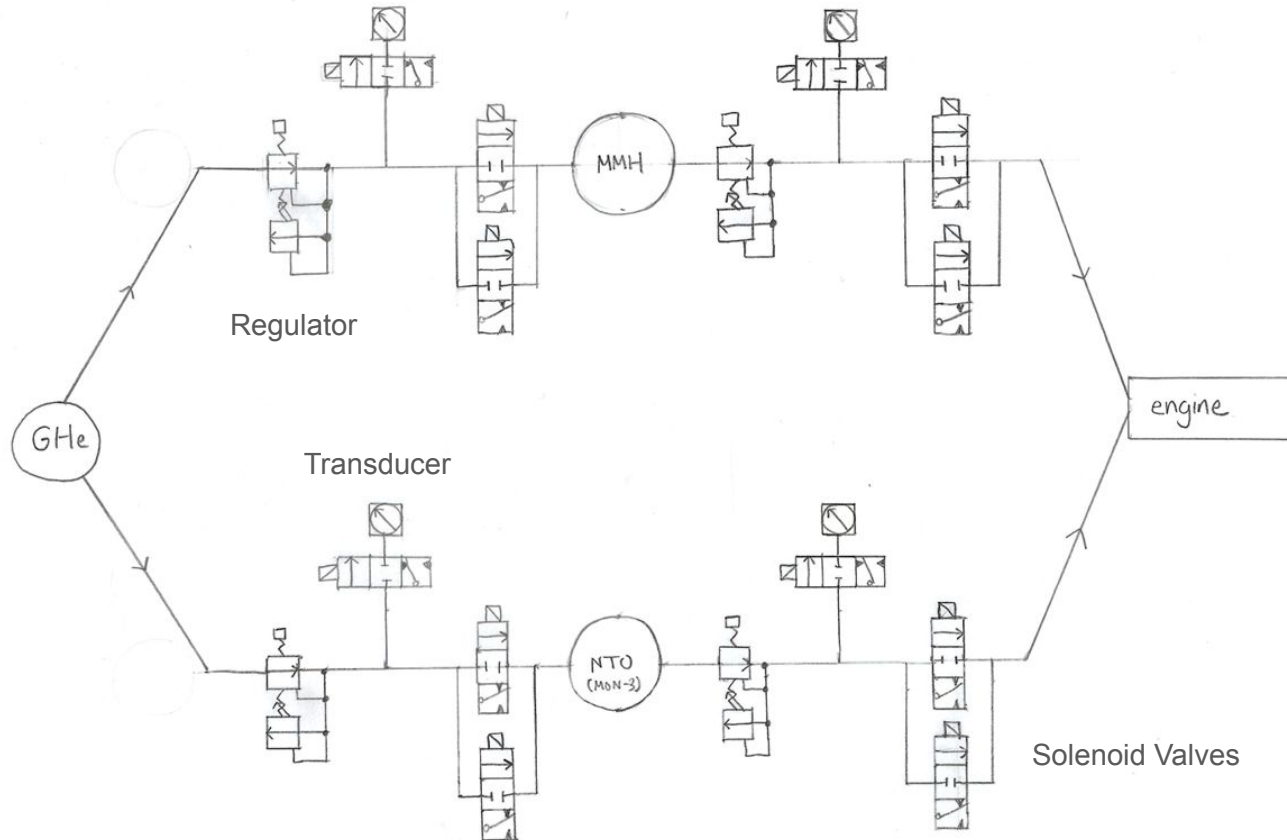
- MMH/NTO (MON-3)
 - Monomethylhydrazine as fuel
 - Nitrogen Tetroxide with 3% Nitric Oxide as the oxidizer
- Mass: 10.5 kg
- Specific Impulse: 293 secs
- Thrust: 4000 Newtons
- Delta V Requirement: 265.20 km/s
- Propellant Mass: 1000 kg
 - Fuel: 384.62 kg
 - Oxidizer: 615.83 kg
- O/F Ratio: 1.6



Propulsion - Main Spacecraft Engine Tank Sizing

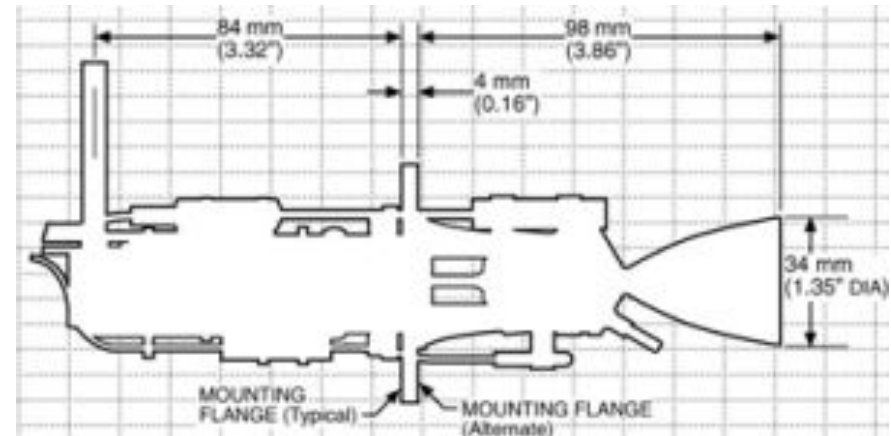
- Selected parameters:
 - Factor of Safety 1.3
 - Aluminium Alloy 7075
- Fuel Tank:
 - Spherical - Radius of 0.47 meters
 - Mass - 7.88 kg
 - Volume - 0.46 meters cubed
- Oxidizer Tank:
 - Spherical - Radius of 0.46 meters
 - Mass - 7.73 kg
 - Volume - 0.45 meters cubed
- Pressurant Gas Tank:
 - Spherical - Radius of 0.28 meters
 - Mass - 29.79 kg
 - Volume - 0.092 meters cubed

Propulsion - Main Spacecraft Engine Diagram



Propulsion - MR-106L Thruster

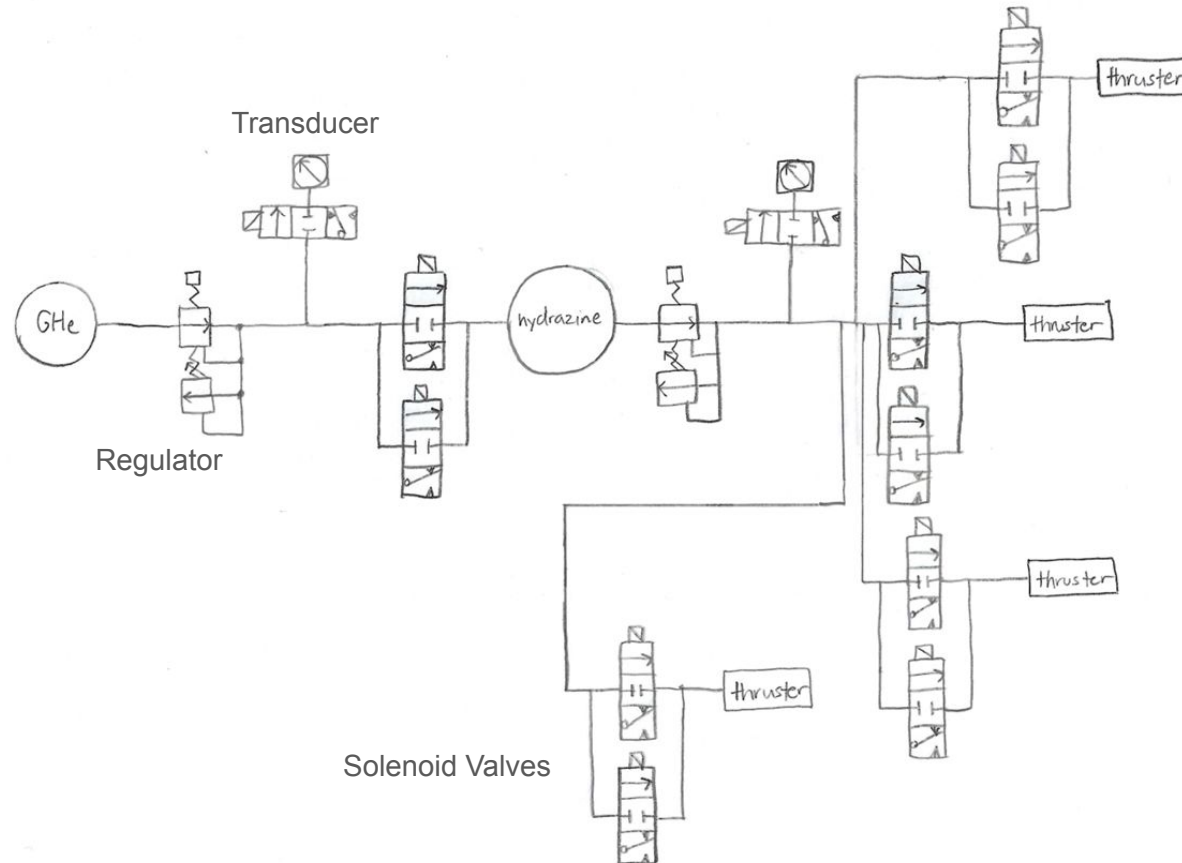
- Propellant: hydrazine
- Configuration: 16 Thrusters
- Mass: 0.59 kg
- Specific Impulse: 230 secs
- Thrust: 22 Newtons
- Delta V Requirement: 10 km/s
- Propellant Mass: 40 kg



Propulsion - RCS Thrusters Tank Sizing

- Selected Parameters:
 - Factor of Safety 1.3
 - Aluminium Alloy 7075
- Propellant Tanks - 4:
 - Spherical - Radius of 0.043 meters
 - Mass - 0.26 kg
 - Volume - 0.01 meters cubed
- Pressurant Gas Tanks - 4:
 - Spherical - Radius of 0.02 meters
 - Mass - 0.17 kg
 - Volume - 0.0005 meters cubed

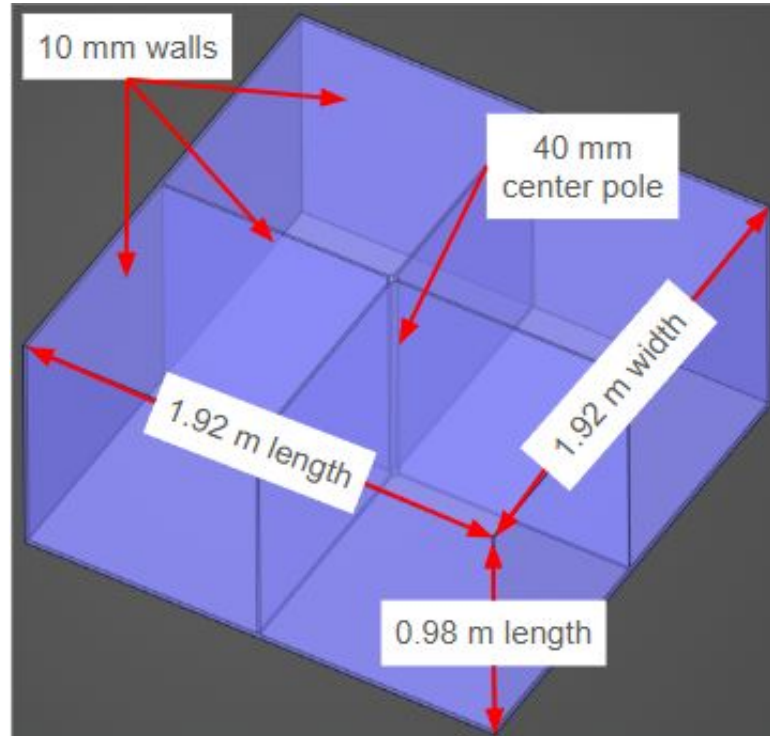
Propulsion - RCS Thrusters Diagram



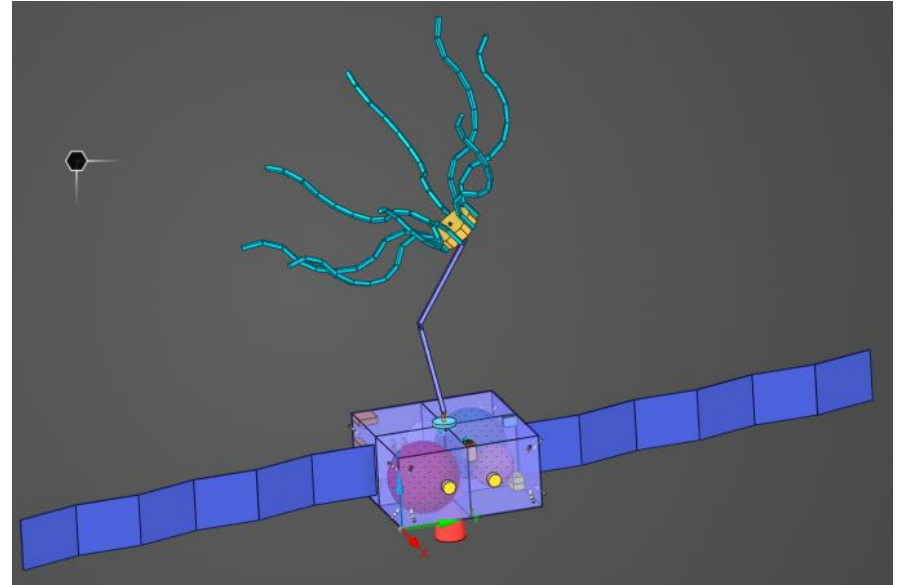
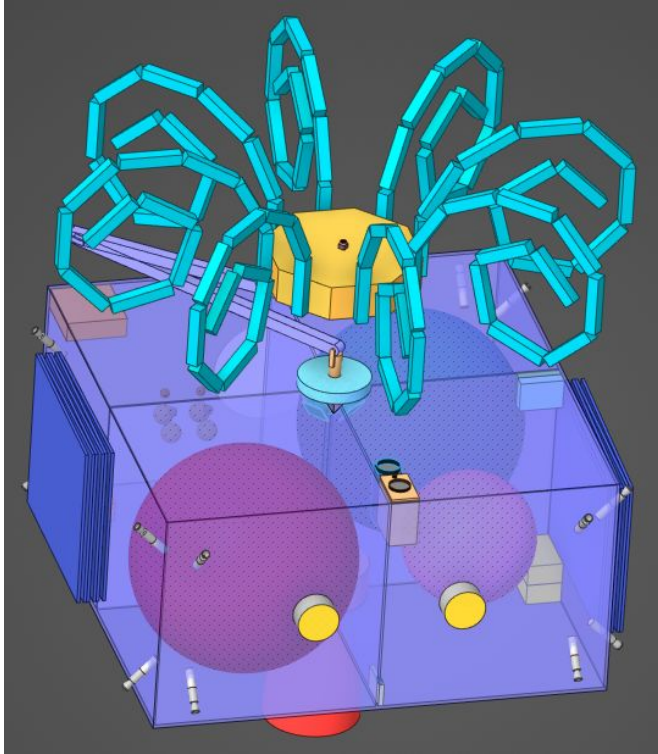
Spacecraft Configuration & Structures - Overview

- Provides the structure on which all spacecraft systems are mounted
- Handles the structural, thermal, and environmental loads experienced during each phase of the mission.
 - The highest structural loads occur during deployment and main engine firing.
- Made of high strength, low mass materials commonly used in the spacecraft industry (Aluminum alloy 6061 T6)

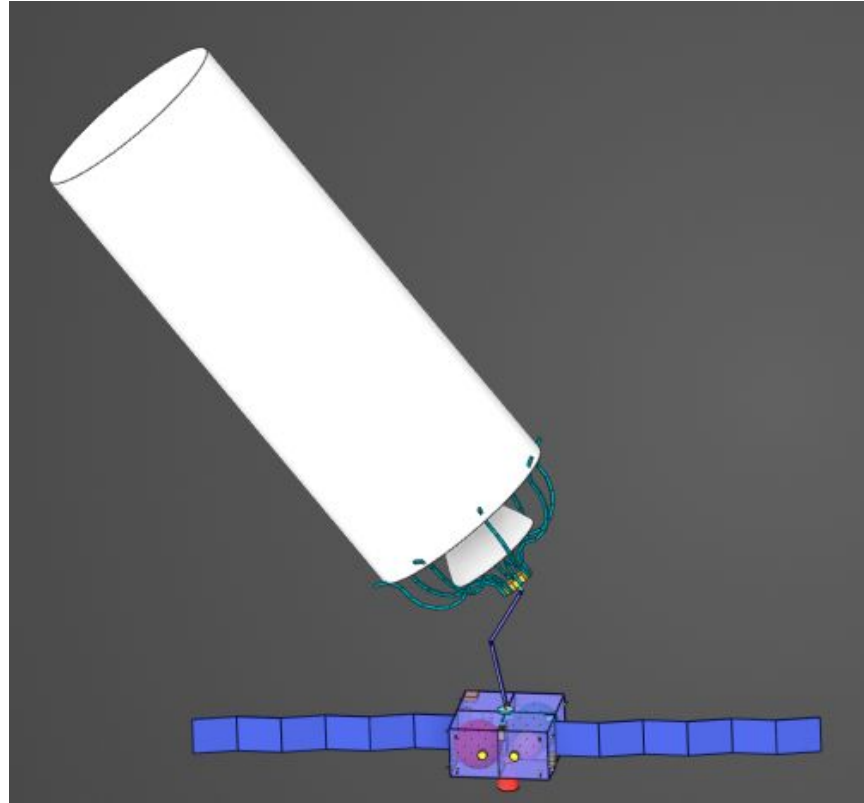
Spacecraft Configuration & Structures - Frame size



Spacecraft Configuration & Structures - undeployed vs deployed



Spacecraft Configuration & Structures - Captured debris

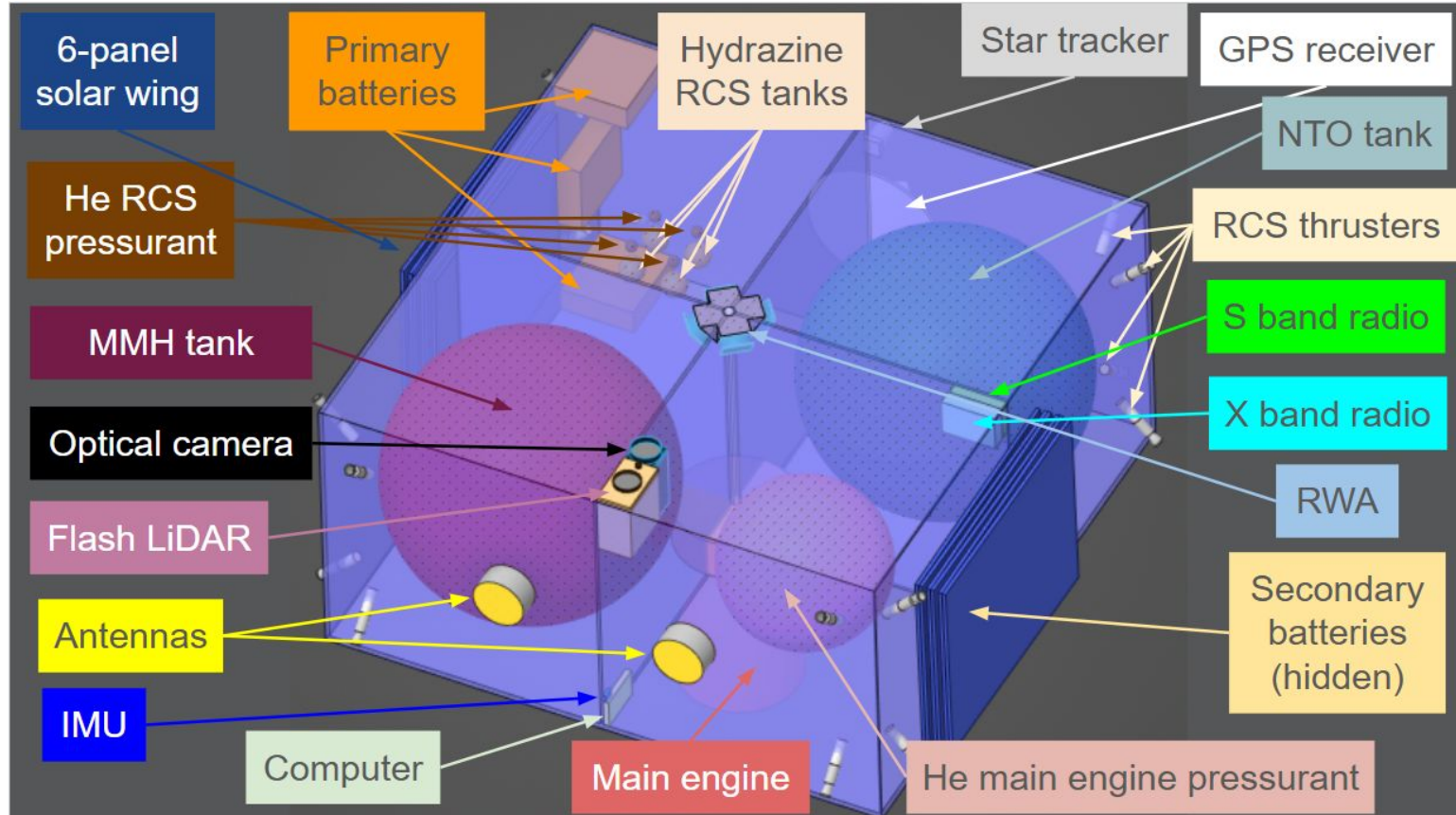


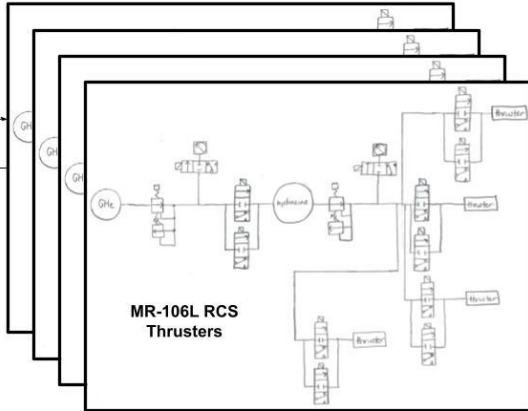
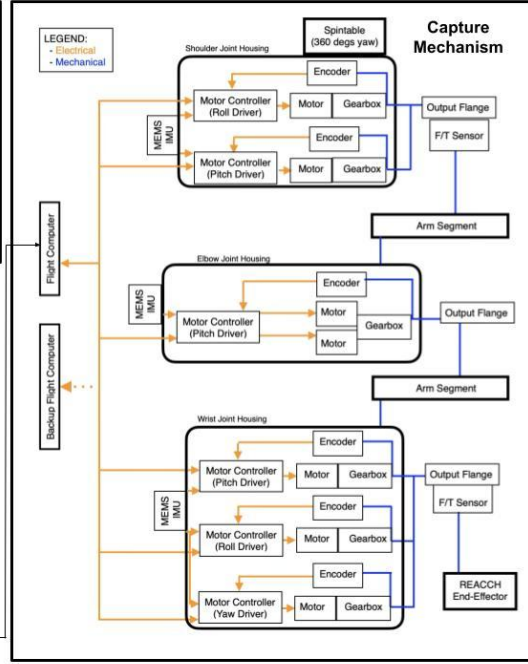
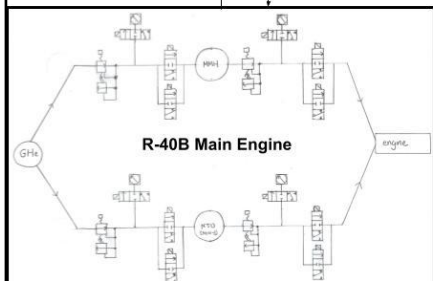
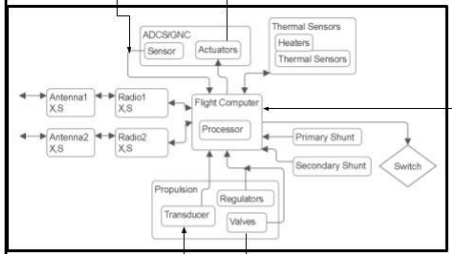
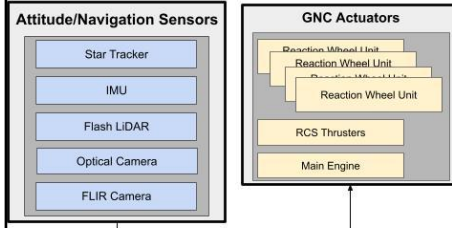
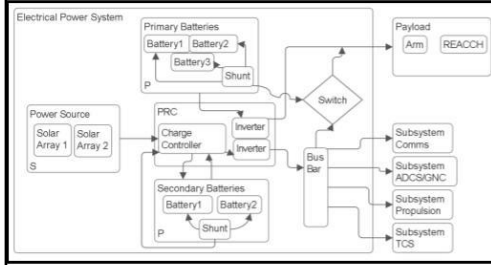
Spacecraft Configuration & Structures - BOM

Part	Part name	Quantity	Vtot (m^3)	Mass (kg)
Structure	Custom Aluminum 4-Quadrant Frame	1	3.61E+00	499.188
REACCH and PFMA	KMI REACCH and PFMA	1	1.12E+00	90
Main Thruster	Redmond R-40B	1	9.00E-02	6.8
RCS thrusters	Aerojet Rocketdyne MR-106L	16	2.64E-03	32
Main prop tank (empty)	Custom Aluminum Sphere	1	4.64E-01	7.869
Main ox tank (empty)	Custom Aluminum Sphere	1	4.51E-01	7.718
Main Engine pressurant (empty)	Custom Aluminum Sphere	1	9.15E-02	28.654
RCS prop tank (empty)	Custom Aluminum Sphere	4	5.03E-02	0.256
RCS pressurant (empty)	Custom Aluminum Sphere	4	3.37E-05	0.148
Solar Panels	Lockheed Martin Small Satellite Solar Array	12	7.52E-03	38.1018
Primary Batteries	Ibeos 50V 1 kWhr Battery	3	1.78E-02	22.5
Secondary Batteries	Ibeos 8S4P 550Whr Battery	2	6.35E-03	8
Flash Lidar	ASC GSFL-16KS	1	5.07E-03	3.2
Optical camera	Infinity Avionics ORION12MP-550	1	2.50E-03	1.5
IR Camera	Teledyne BLACKFLY S BFS-U3-13Y3	1	2.52E-05	0.036
RWUs	Space Inventor Reaction Wheel WHL-500mNms	4	1.51E-03	3.2
IMU	HBK 3DM-CX5-IMU	1	9.60E-06	0.008
Star tracker	Terma T3 Star Tracker	1	3.60E-04	0.35
GPS reciever	L3Harris AS-48917 GPS BAND OMNIDIRECTIONAL ANTENNA	1	3.13E-03	0.454
X Radio	RocketLab Frontier-X	1	1.18E-03	0.92
S Radio	RocketLab Frontier-S	1	7.90E-04	0.59
Antenna	Flexitech 650MHz-10GHz Spiral Antenna	2	4.00E-03	2.566
Computer	FrontGrade Generation 8 (GR740 Based) SBC	1	3.20E-04	0.65

758 kg
Dry Mass

Spacecraft Configuration & Structures - Component layout





Spacecraft Mechanical - Capture Mechanism Overview

- Key Functions

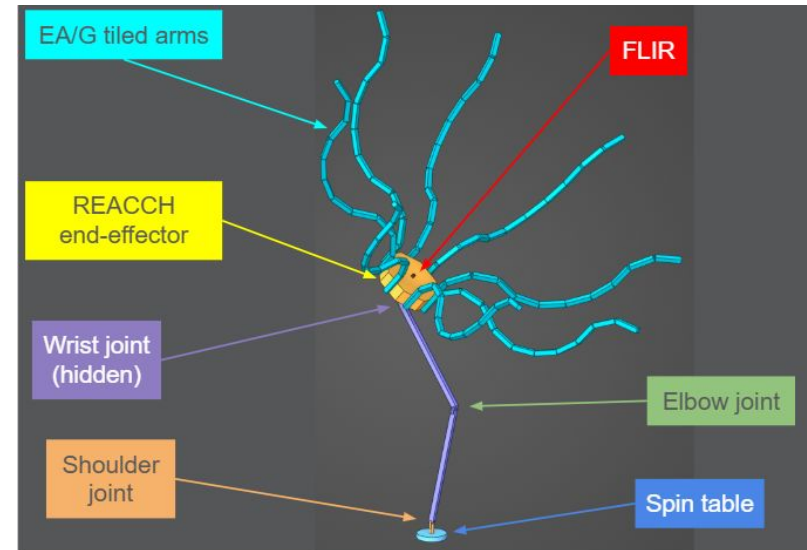
- Enables autonomous capture and stabilization of an uncooperative target object
- Stabilizes PERSEUS/target object pair post-capture for a safe controlled deorbit
- Major Components:
 - 2.4 m robotic arm
 - 8-tentacle REACCH end-effector
- Designed for high reliability during proximity operations and debris-capture maneuvers

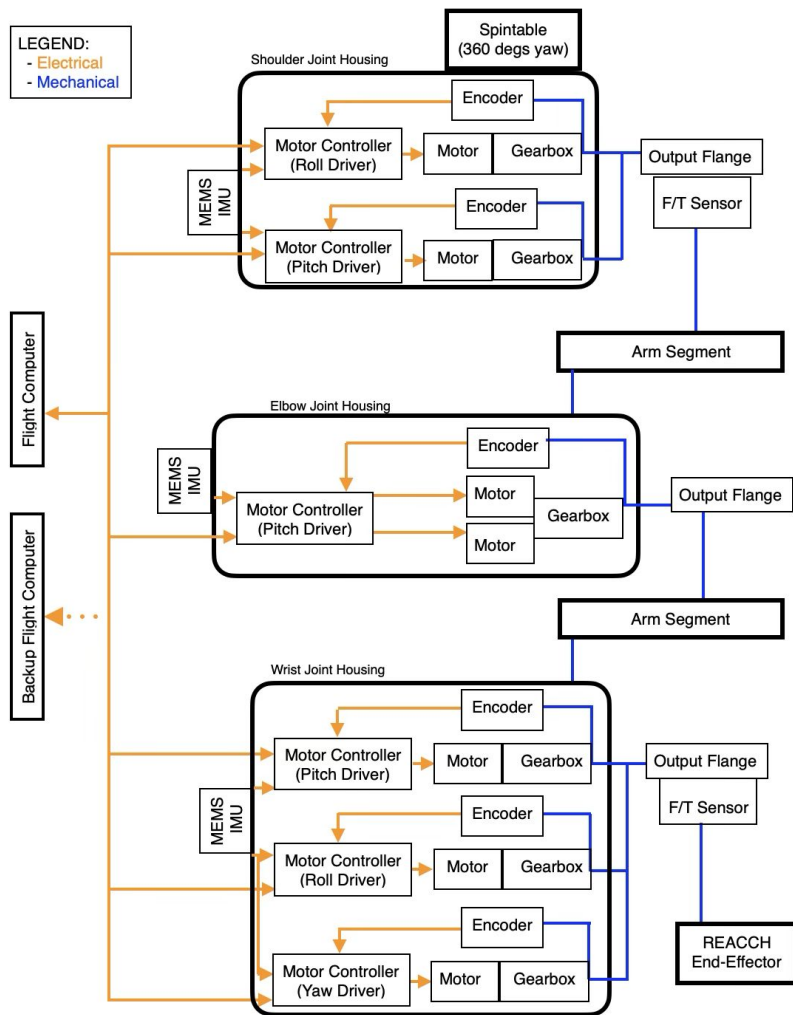
- Structural Interface

- Mounted via torsionally resistant isogrid ring to align loads to spacecraft thrust structure
- Hold-Down & Release Mechanism (HDRM) ensures launch restraint and low-shock arm deployment

Spacecraft Mechanical - Robotic Arm

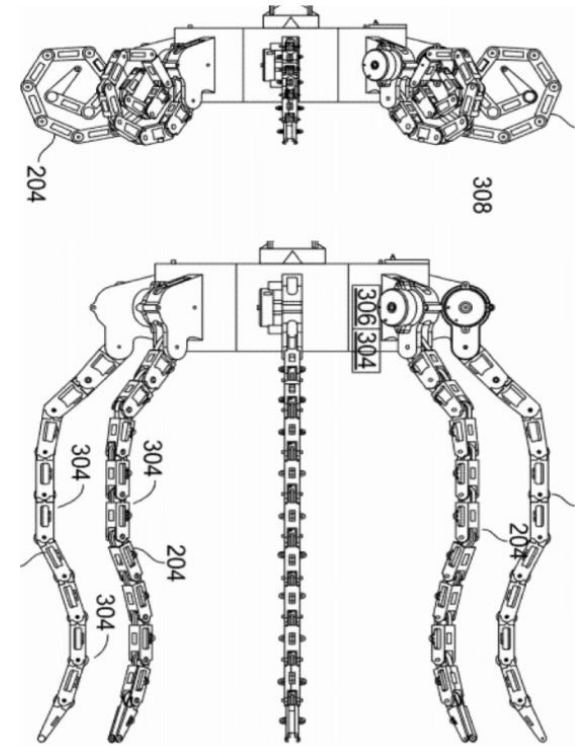
- Heritage & Configuration
 - Derived from Shuttle-era P-FMA; upgraded with modern actuators, sensors, & avionics
 - Provides 7 Degrees of Freedom
 - Shoulder Pitch/Yaw/Roll
 - Elbow Pitch
 - Wrist Pitch/Yaw/Roll
- Materials & Mass
 - Shoulder & Elbow: Ti-6Al-4V for high strength and fatigue resistance
 - Arm Segments and Wrist: CFRP for stiffness and low mass
 - Total Mass (including REACCH): ~86 kg





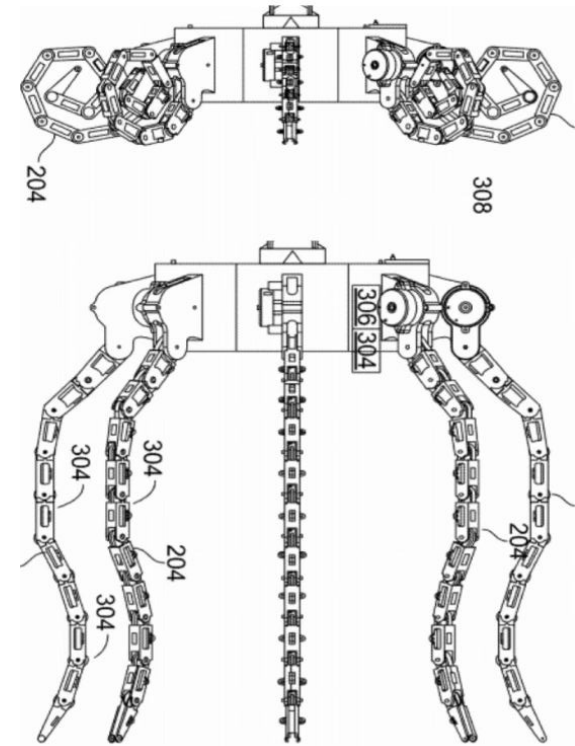
Spacecraft Mechanical - REACCH Capture Mechanism

- Capabilities
 - Eight 2-meter rigid-linked tentacles
 - Capture Range: 25 cm $\rightarrow$ >6.5 m diameter
- Hybrid Adhesion
 - EA tiles: controllable adhesion to varied surfaces
 - Gecko-like microstructures for rough-surface grip
 - Hybrid EA/G tiles dramatically increases shear force & softens impact loads
- Reliability
 - 1000 ground tests, 172 on-orbit captures
 - 2024 - ISS demo
 - TRL - 7



Spacecraft Mechanical - REACCH Mechanical Design

- Tentacle Structure
 - Rigid-segmented spine with low-friction ball bearings
 - Internal pulley for extension/retraction
 - Provides high-load path from tiles to end-effector base
- Sensing & Imaging
 - Each tentacle includes:
 - Power/data lines
 - EA/G tile control
 - Stretch-load sensors
 - Teledyne FLIR camera provides position data that is fused with LiDAR data for precision targeting



Spacecraft Mechanical - Dynamic Capture Operations

- Pre-Capture Alignment
 - Arm adjusts elbow to match target precession (up to $\sim 20^\circ$)
 - Spin-up via Moog UPT3 gimbal (i.e. spin table) to reduce relative motion < 0.25 m/s
- Compliant Contact
 - Wrist integrated translational & rotational compliance
 - Allows soft contact, absorbs off-nominal errors, and gradually transfers angular momentum
 - Full stabilization achieved in ~ 30 - 35 seconds, after which joints lock for rigid control

Thermal Control System - Overview

- The Thermal Control System (TCS) ensures spacecraft components remain within safe temperature limits throughout the mission
- Operates in a vacuum environment - heat transfer mainly via radiation
- Ensures performance during all mission phases
 - Launch, orbit, transfer, NMC, rendezvous, capture, deorbit
- Balance internal heat from electronics with external sources
 - External sources: Sun, Earth, deep space
- Minimize power consumption and mass
- Handle environmental variations
 - Eclipse, attitude changes

Thermal Control - Objectives

- Balance internal heat from electronics with external sources
 - External sources: Sun, Earth, deep space
- Minimize power consumption and mass
- Handle environmental variations
 - Eclipse, attitude changes

Thermal Control - Temperature Requirements

- Internal spacecraft temperature: $15^{\circ}\text{C} \pm 15^{\circ}\text{C}$
- Hull radiates $\sim 5,200 \text{ W}$ at optimal temperature (15°C)
- Orientation affects heat absorption:
 - Ram: -770.77 W
 - Zenith: $-1,206 \text{ W}$
 - 45° angle: $-1,447 \text{ W}$

Component	Min temp C	Max temp (C)	Min temp (K)	Max temp (K)
Robot arm	-35	85	238.15	358.15
Star Tracker	-40	30	233.15	303.15
IMU	-40	85	233.15	358.15
GPS reciever	-100	100	173.15	373.15
Optical Camera	-10	50	263.15	323.15
Flash LiDAR	0	30	273.15	303.15
FLIR	0	50	273.15	323.15
Reaction wheels	-20	60	253.15	333.15
Fuel tank	-50	88	223.15	361.15
Oxidizer tank	-30	50	243.15	323.15
Pressurant tanks	0	30	273.15	303.15
Secondary Battery	0	45	273.15	318.15
Primary Battery	0	45	273.15	318.15
Antenna Array	-20	85	253.15	358.15
X Radio	-25	55	248.15	328.15
S Radio	-30	60	243.15	333.15
Computer	-25	65	248.15	338.15

Thermal Control - Methods

- Two primary methods of thermal control
- Passive:
 - MLI, coatings, passive heat pipes
- Active:
 - Electric thermofoil heaters

Thermal Control - Passive

Passive Elements

- Aeroglaze A276 coating
 - Low absorptivity, high emissivity
- Heat pipes with ammonia
 - $T_b = 240 \text{ K}$
 - Transfers thermal energy from hot spots to frame to radiate excess heat to space

Thermal Control - Active

Minco HK6907 Thermofoil heaters

- Used during startup, eclipse, preconditioning
- Controlled by thermal management system



Thermal Control - Cooling System

Radiative Cooling from Hull

- Hull acts as a radiator, Area = $\sim 14.89 \text{ m}^2$
- Absorptivity = 0.23, Emissivity = 0.88 - 0.91 (from coating)
- Rejects $\sim 5.2 \text{ kW}$ under steady-state conditions

Thermal Control - Heating System

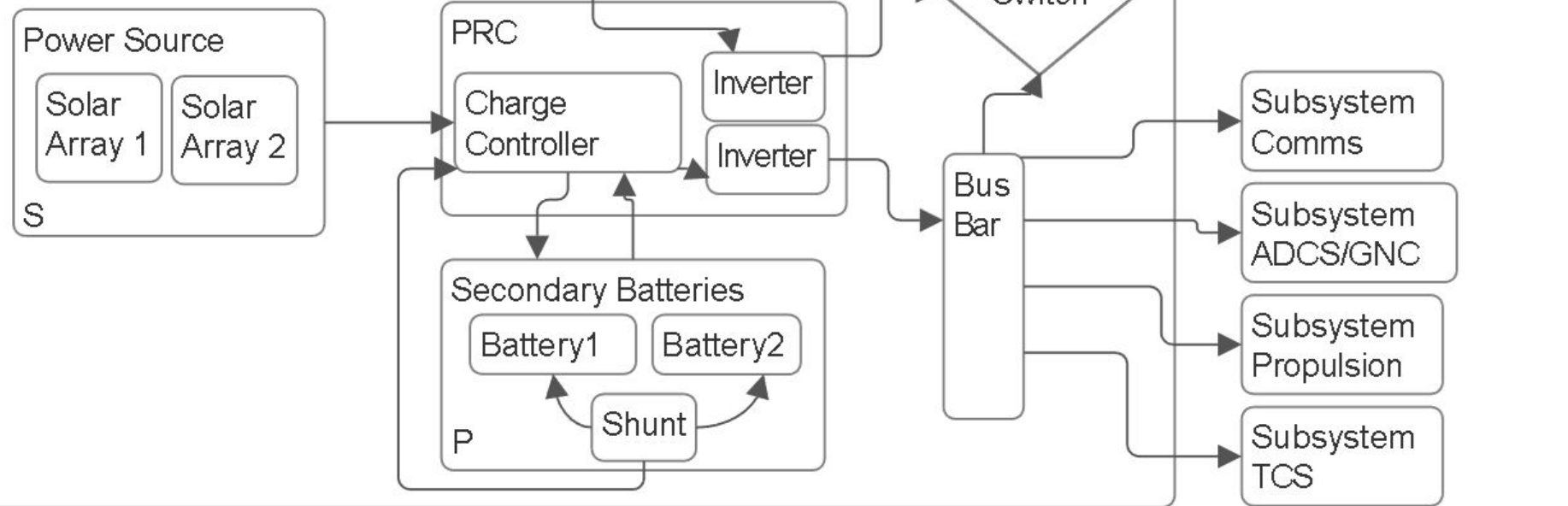
Minco Heaters

- Installed on surfaces such as: battery plates, avionics, and propellant tubing
- Adhesive-backed fastening provides effective thermal contact
- Construction is lightweight and low outgassing
- Heaters fitted with temperature sensors controlled by the thermal management system

Power Systems - Overview

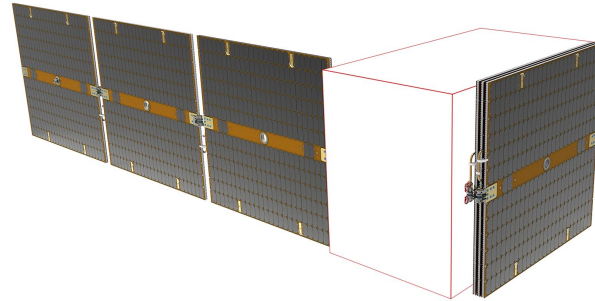
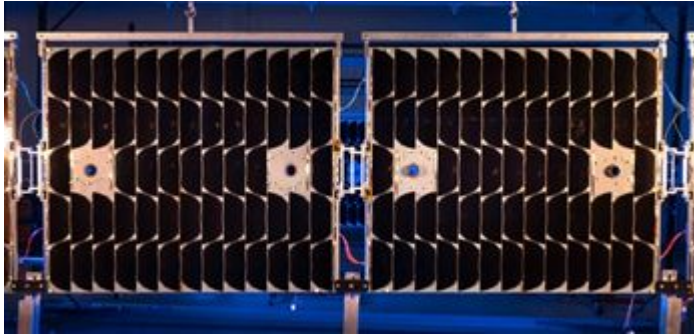
- Generate power for multiple modes (Solar Panels)
- 230.8 Watts for low power mode
- 809.36 Watts for maneuver mode
- 1381.36 Watts for deployment phased mode
- 2 Rechargeable 8S4P secondary batteries
- 3 1000 Whr primary batteries dedicated for Arm and REACCH

Electrical Power System



13 - Lockheed Martin Small Satellite Solar Array

- Primary Solar Array
 - Gallium Arsenide Multi-junction
 - 170 W BOL - $\sim 295.7 \text{ W/m}^2$
- 12 Panels to supply power

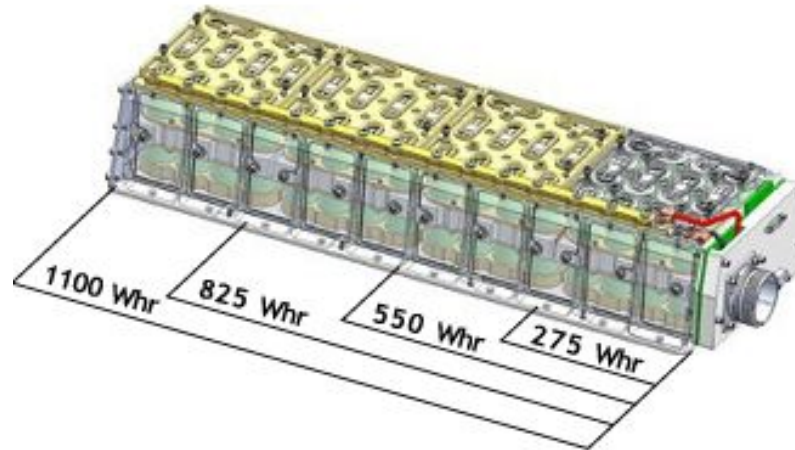


Solar Panel Sizing

- Power Required 1381.36 Watts
- Orbital Period 101.799 min (1.697 hr)
- Daylight time 66.722 min (1.112 hr)
- Eclipse time 35.077 min (0.585 hr)
- 2343.7 Whr per orbit needed
- 2107.57 Watts needed during daylight.
- Total area needed: 7.127 m² of charging surface
- 12 Panels needed

2 - Ibeos 28V Modular Secondary Batteries

- Two Configurations selected
 - 2-550 Whr (for redundancy)
- Total mass: 8 kg



3 - Ibeos 50V/kW hr Primary Batteries

- Dedicated for Payload power
- Total mass: 22.5 kg
- Total 3000 Whr

Needed: 2600 Whr



Battery Sizing

Based on Power Usage during Payload phase - 1381.36 Watts

Eclipse Period - 35.077 mins (0.585 hr)

Whrs to work through the Eclipse period = 807.5736

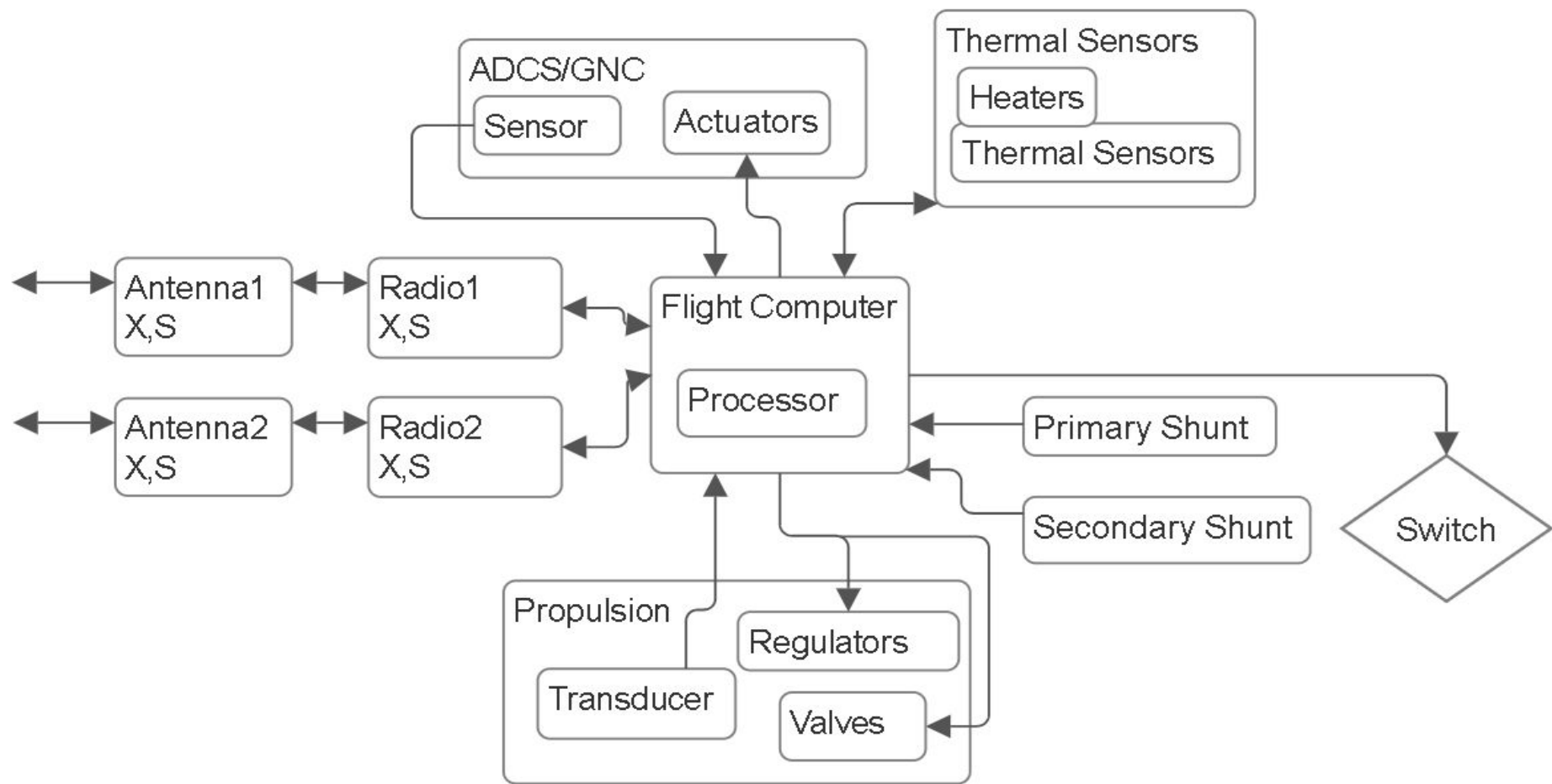
Total battery provided Whrs = 1100

Depth of Discharge: 73.42%

Industry standard DoDs: 40-80% with longer missions having lower DoDs

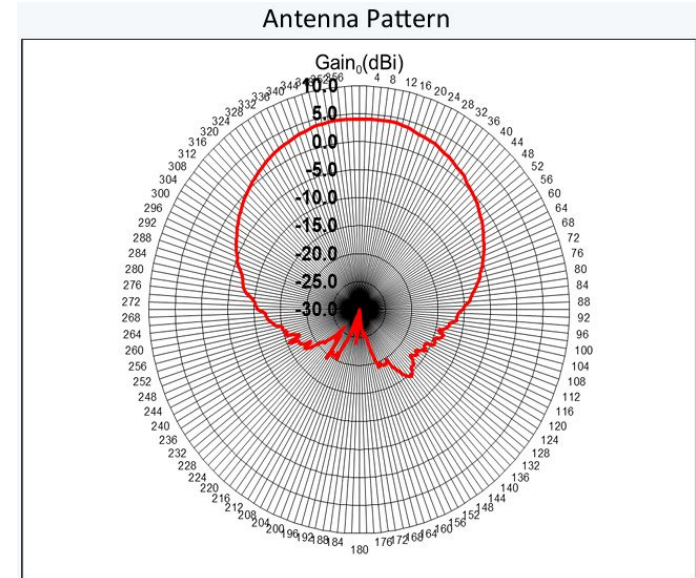
Communications and Data Systems Overview

- Provides continuous uplink and downlink between the spacecraft and Near Earth Network Ground Stations
 - Large coverage area
- 2 X,S band antenna
- 2 X,S band radios
- 1 GPS receiver dedicated for ADCS subsystem
- GR740 Quad-Core LEON4FT Processor (best compatibility with payload)



2 - Flexitech Aerospace Spiral Antennas 650 MHz-10 GHz

- Uplink Frequencies; 2.07 GHz in S-Band, 7.19 GHz in X-Band
- Downlink Frequencies; 2.25 GHz in S-Band, 8.45 GHz in X-Band
- Commercially Available
- 10 Watts each
- Boresight Gain: 3 dBi
- Mass: 1.283 kg
- Pointing Accuracy:
 - 1dB at +/-34 deg.
 - 3dB at +/-60 deg



As-48917 GPS Band Omnidirectional Antenna

- L1 (civilian) and L2 (encrypted military) GPS frequencies
- Frequency 1.217 - 1.585 GHz
- 3dB Beamwidth 110 deg min,
- Power Handling: Receive only
- Gains
 - 0 dBi +/-10 deg
 - -5 dBi +/-76 deg
- Dedicated to ADCS

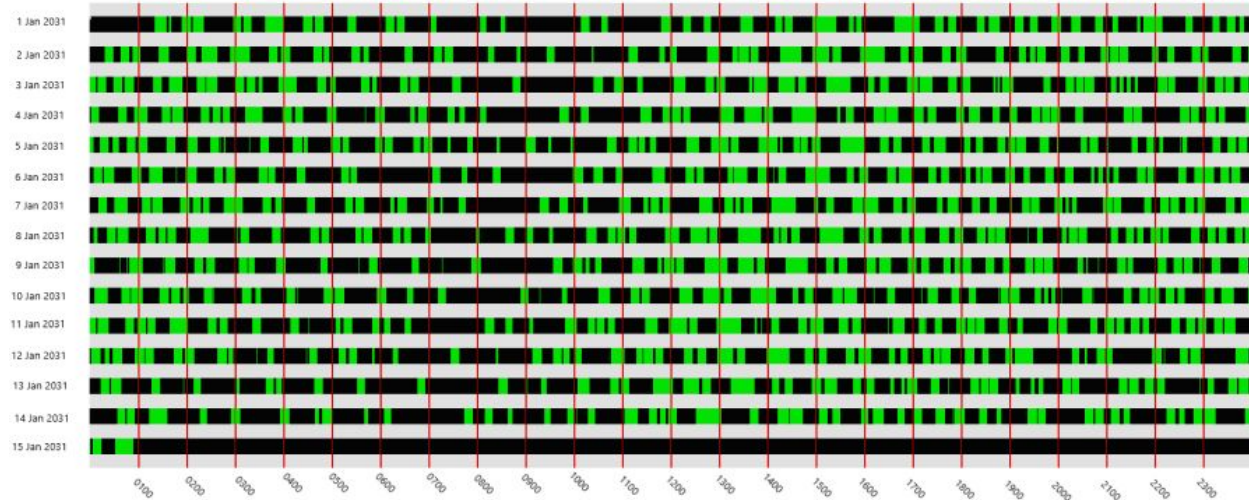


2 Frontier Radios for S and X Bands and Spacewire

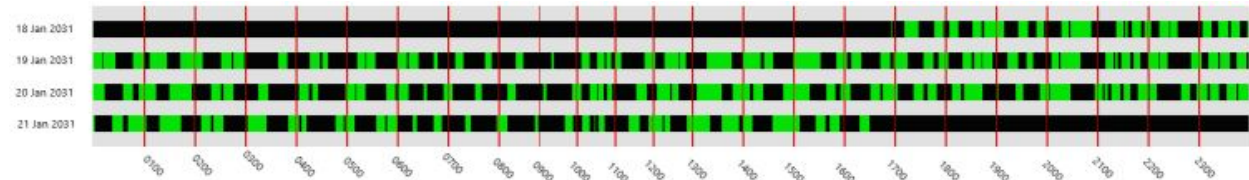
- Connects with the Near Earth Network
- Provides information for Telemetry, Tracking, and Command
- S uplink range 2020 - 2120 MHz
- S downlink range 2200 - 2300 MHz
- X uplink range 7145 - 7235 MHz
- X downlink range 8400 - 8500 MHz
- Radios connect with Spacewire standards Processor
- Using Spacewire Standard Network for data handling within PERSEUS
- Available Modulations:
 - BPSK Carrier wave shifts by 180 degrees, lower data rates
 - QPSK Carrier wave shifts by 90 degrees, higher data rates

Communications blackout periods

Before
payload capture



After
Payload capture



Link Budget for Fairbanks Ground Station. 11.28 m dish

	S Band	X Band
Available Uplink (dB)	241.4403	249.2327
Available Downlink (dB)	21.45768	19.41745
End to End Eb/Co (dB)	262.898	268.6502
Required Eb/Co (dB)	15.2042	15.52042
Link Margin (dB)	247.3776	253.1297

Reliability and Risk

- By looking at the worst-case scenarios encountered by each subsystem, a list of top-level risks is formed
- By considering the likelihood and impact of these scenarios occurring, a qualitative risk assessment can be performed

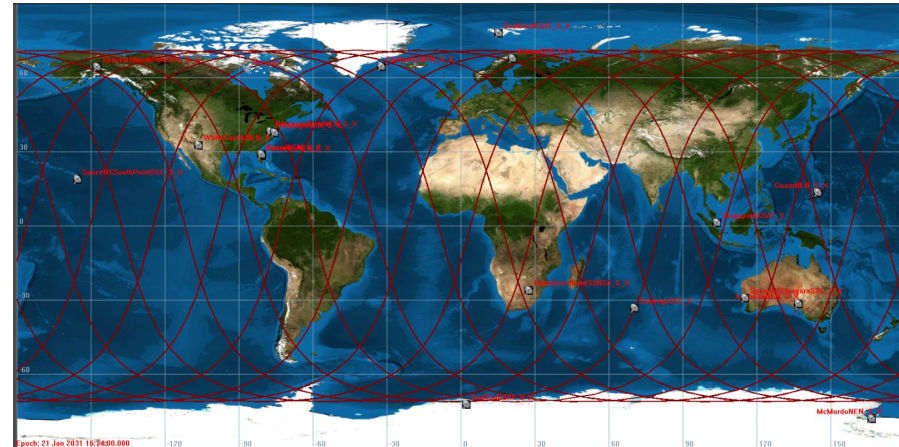
	Source	Risk considerations	Risk Level
R-1	Power: Battery failure	With two medium capacity secondary batteries and three high capacity primary batteries, PERSEUS can operate with next to no interruption with single failures in either type of battery	Unlikely, low impact
R-2	Power: Solar Array deployment	Having two Solar Array wings allows PERSEUS to still generate power if deployment fails. The presence of primary batteries independent from the Solar Array drastically reduces the impact of a deployment failure. Solar Array deployment historically has a comparatively high failure rate	Unlikely, low impact
R-3	Communications: Pointing failure leads to loss of communications	There are two separate omni-directional antennas, neither are independently pointable	Unlikely, high impact
R-4	Data: Computer failure	Computer failure is unlikely, but would have a significant impact on mission success	Unlikely, medium impact
R-5	GNC: Sensor/reaction wheel failure	With a high heritage design and a short mission duration, failure is very unlikely. With multiple navigation sensors and two different pointing methods, impact would be somewhat low	Unlikely, somewhat low impact
R-6	Propulsion: Thruster/fuel failure	With built-in fuel margin and high-heritage thrusters, likelihood of failure is low, but the impact would be very high	Very unlikely, high impact
R-7	Environment: Single Event Effects (SEE)	Radiation tolerant design and a solar-minimum mission window mitigates the constant problems associated with radiation	Very likely, low impact
R-8	Capture: Breakoff of more debris	With an uncooperative docking scenario, even with careful capture mechanism operation, there is a possibility of overstressing the aged target. This may damage PERSEUS, and the goal of the mission is debris mitigation.	Unlikely, high impact
R-9	Astro/GNC: Can't characterize target motion	It isn't likely that tracking would fail to characterize the target's motion due to a high number of targeting sensors, but due to the delicate nature of the capture process, this could have a significant effect on mission success	Very unlikely, somewhat high impact
R-10	Astro: Collision risk	Once PERSEUS is added to the SCC catalog, mission operators will receive collision notifications, giving them opportunity to react	Very unlikely, high impact

Very Likely	R-7				
Somewhat Likely					
Moderately Likely					
Unlikely	R-1, R-2	R-5	R-4		R-3, R-8
Very Unlikely				R-9	R-6, R-10
	Low Impact	Somewhat Low Impact	Medium Impact	Somewhat High Impact	High Impact

	Monitor
	Manage Closely
	Redesign

Launch and Mission Operations

- Launching from Vandenberg
- Ground operations out of Ames Research Center
- NASA's Near Earth Network and Commercial sites
- 6 crews
 - 3 members per crew: Mission Commander, Satellite Vehicle Operator, Orbit Analyst
 - Weekdays - 3x 8 hour shifts
 - Weekend - 2x 12 hour shifts
 - 5 rotating crews with 1 backup crew



Cost - Estimation Method

- To make an estimate of the total cost of the PERSEUS mission, we chose to use the NASA QuickCost estimation method shown in The New SMAD.
- It has a standard error of 41%, meaning this estimate is preliminary, but still gives a helpful range of possible costs.
- Variables contributing to the QuickCost chart were estimated based on the final designed state of each of PERSEUS' subsystems.

Cost Driver	Value	Description	Rationale
Dry mass	758	Dry mass of spacecraft and instruments in kg	Current estimate for SC dry mass
Power	1360	LEO equivalent Beginning Of Life power (BOLP) in Watts	Current estimate for SC BOLP
Data%	0.25	Data Rate Percentile (fraction) relative to the state-of-the-art at Authority To Proceed (ATP) (0.5 for median data rate missions, <0.5 for lower data rates, >0.5 for higher data rates)	Data rate is expected to be low for this mission due to a lack of scientific objectives
Life	0.75	Advertised Design Life (in months) including extended operations	Current estimate for SC design life
New	0.3	Percent New (fraction): 0.2-0.3 = Simple Mod, 0.3-0.7 = Extensive Mod, 0.7-1.0 = New	Most of the spacecraft will be made up of COTS parts, but the capture mechanism makes use of a combination of TRL 7 components that are as of yet untested together
Planetary?	0	0 for Earth orbital, 1 for planetary	Earth orbital
Year	2027	ATP date in 4 digit calendar year minus 1960	Phase B is expected to begin in 2027 for our expected launch date
InstrComp%	0.5	Instrument complexity percentile (fraction) - relative to "average" instrument complexity (enter 0.5 for median complexity, <0.5 for lower complexity, >0.5 for higher complexity)	The capture mechanism has a medium level of complexity
Team	4	Team Experience: 1 = Unfamiliar, 2 = Mixed, 3 = Normal, 4 = Extensive	The team will be made up of members with an extensive level of experience

Cost - Additional Adjustments

Consideration	Percent increase (%)	Cost increase (2025 millions)
Ground station retrofit	9	7.428698184
Mission Operations and Data Analysis	10	8.254109094
Launch	N/A	70
Total	19	85.68280728

Cost - Final Estimate

Cost estimate: 168.224 in 2025 millions.

Cost range with error: 127.952 - 208.496 in 2025 millions.

Cost - Comparison

- The estimated cost of a similar ESA-led debris removal mission, Clearspace-1, has a cost of 156.646 in 2025 millions (134.816 in 2025 million euros)
 - This is very close to the PERSEUS estimate of 168.224 in 2025 millions.
- The mission of Clearspace-1 involves handling debris in a 80km lower orbit with a smaller profile and 800 kg lower mass than the target of the PERSEUS mission, so an increase in a few million dollars is expected.

Summary

- Goal: Conduct close proximity operations to capture a large piece of space debris and complete a controlled re-entry
- Target Object: Zenit-2 Upper Stage
- Outcome: Capture and Controlled Reentry

Summary - System Integration Complete

- All major subsystems defined
- Final spacecraft configuration completed
- Subsystems verified for compatibility

Summary - Operations and Cost

- Launch Site: Vandenberg
- Network: Near Earth Network
 - Rotating Ground Crews
- Cost:
 - Estimation: \$168,224,000

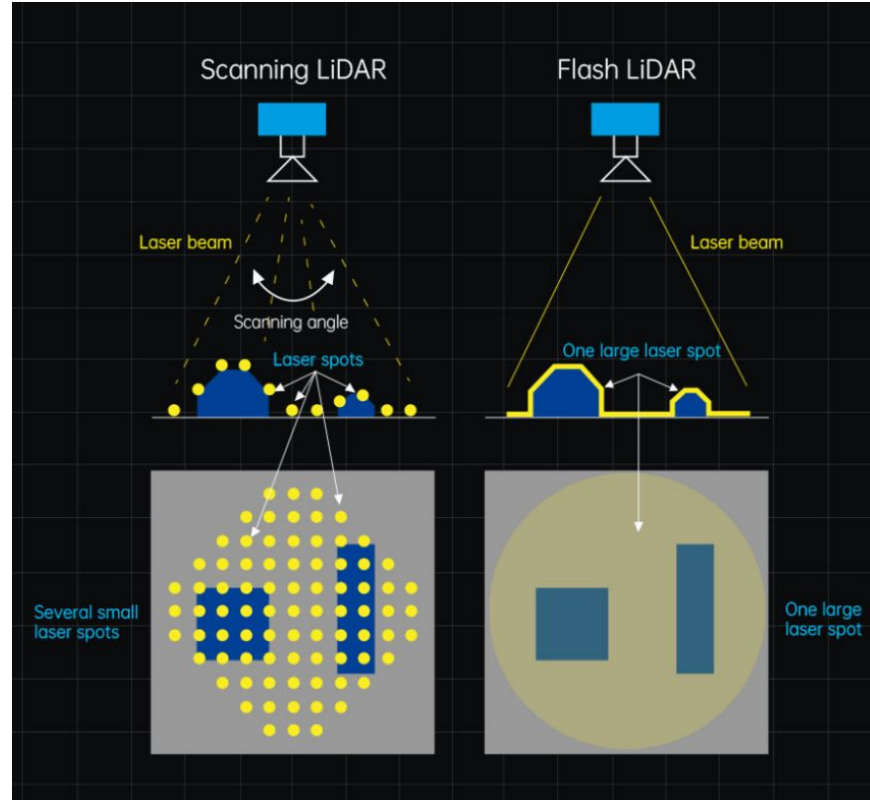
Conclusion

- This mission is important because space debris is a major concern for space exploration.
- 17 of these boosters are on the “Top 50” list
 - They are extremely dangerous
 - Top 50 statistically most concerning derelict objects in space list

#	Object Name/SATNO	Risk, kg	Count	Mass, kg	Ave Alt, km	Incl, deg	Launch
1	28353 (SL-16 R/B)	283	341	9000	844	71.0	2004
2	27386 (ENVISAT)	175	276	8211	762	98.3	2002
3	7727 (COSMOS 724)	135	25	3769	897	65.6	1975
4	19120 (SL-16 R/B)	117	366	9000	827	71.0	1988
5	24279 (H-2 R/B)	115	133	2700	1082	98.7	1996
6	39203 (CZ-2C R/B)	107	142	4000	705	98.4	2013
7	22565 (COSMOS 2237)	106	64	3200	851	70.8	1993
8	5917 (METEOR 1-11)	94	33	1200	862	81.2	1972
9	16292 (SL-8 R/B)	86	33	1435	974	82.9	1985
10	28060 (SERVIS 1)	86	11	840	992	99.5	2003
11	19650 (SL-16 R/B)	71	262	9000	839	71.0	1988
12	15986 (COSMOS 1677)	60	19	3769	940	64.7	1985
13	36123 (CZ-4C R/B)	60	37	2000	1086	100.5	2009
14	22823 (SPOT 3)	59	67	1869	826	98.9	1993
15	24304 (COSMOS 2334)	59	16	820	986	82.9	1996
16	44548 (CZ-2D R/B)	55	225	4000	757	98.3	2019
17	16511 (SL-8 R/B)	52	51	1435	978	82.9	1986
18	25590 (COSMOS 2361)	51	12	820	988	82.9	1998
19	11511 (SL-8 R/B)	50	89	1435	765	74.0	1979
20	20625 (SL-16 R/B)	48	344	9000	843	71.0	1990
21	23603 (COSMOS 2315)	47	14	820	989	82.9	1995
22	25407 (SL-16 R/B)	46	323	9000	839	71.0	1998
23	24298 (SL-16 R/B)	44	280	9000	851	70.8	1996
24	6019 (COSMOS 489)	43	13	820	978	74.0	1972
25	8344 (SL-8 R/B)	43	117	1435	758	74.1	1975
26	25567 (NADEZHDA 5)	43	17	825	992	82.9	1998
27	25400 (SL-16 R/B)	40	331	9000	805	98.8	1998
28	17590 (SL-16 R/B)	40	348	9000	835	71.0	1987
29	16012 (SL-8 R/B)	40	83	1435	767	74.1	1985
30	39261 (CZ-4C R/B)	33	168	2000	779	98.9	2013
31	23774 (SL-8 R/B)	33	63	1435	975	83.0	1996
32	22566 (SL-16 R/B)	30	293	9000	842	71.0	1993
33	22220 (SL-16 R/B)	30	330	9000	836	71.0	1992
34	24773 (SL-8 R/B)	29	45	1435	986	82.9	1997
35	16182 (SL-16 R/B)	28	359	9000	837	71.0	1985
36	23705 (SL-16 R/B)	28	331	9000	841	71.0	1995
37	23405 (SL-16 R/B)	27	364	9000	841	71.0	1994
38	10531 (COSMOS 970)	27	13	2000	1036	65.9	1977
39	22803 (SL-16 R/B)	26	341	9000	835	71.0	1993
40	57831 (CZ-6A R/B)	25	12	5800	801	86.0	2023
41	39014 (CZ-4C R/B)	25	71	2000	995	63.4	2012
42	28480 (CZ-2C R/B)	25	135	4000	803	98.2	2004
43	41858 (CZ-2D R/B)	24	242	4000	769	98.6	2016
44	17974 (SL-16 R/B)	22	354	9000	834	71.0	1987
45	26070 (SL-16 R/B)	22	298	9000	840	71.0	2000
46	22285 (SL-16 R/B)	22	341	9000	841	71.0	1992
47	21088 (SL-8 R/B)	21	42	1435	973	82.9	1991
48	13114 (SL-14 R/B)	20	36	1407	947	82.5	1982
49	12319 (COSMOS 1249)	20	27	3769	940	65.0	1981
50	4589 (SL-8 R/B)	19	95	1435	749	74.1	1970

EXTRAS

GNC - Flash LiDAR





0117 UTC
01 Jan 2031
Launch

0127 UTC
15 Jan 2031
Hohman
Transfer

1408 UTC
18 Jan 2031
Ingress



1634 UTC
21 Jan 2031
Deorbit Burn

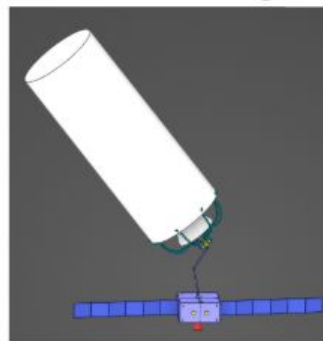
1721 UTC
21 Jan 2031
Splashdown

0127 UTC
01 Jan 2031
Parking Orbit
insertion



0208 UTC
15 Jan 2031
NMC
insertion

1634 UTC
18 Jan 2031
Docking



1714 UTC
21 Jan 2031
Reenter
Earth's
Atmosphere